DPC-GNN-Acoustic

Core Innovation & Mathematical Framework

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Central thesis: *"Physics is not a loss function. Physics is the forward pass."*

1. The Physics-as-Forward Paradigm

Definition 1 (Physics-as-Forward Neural Network)

A neural network $\mathcal{M}_\theta$ satisfies the *Physics-as-Forward* (PaF) property w.r.t. governing equation $\mathcal{L}[u] = 0$ if:

$$\mathcal{M}_\theta = \mathcal{D} \circ \mathcal{P}_\mathcal{L} \circ \mathcal{E}_\theta$$

where $\mathcal{E}_\theta$ is a **learnable encoder** (inputs → constitutive parameters), $\mathcal{P}_\mathcal{L}$ is a **deterministic physics solver** (0 learnable params), and $\mathcal{D}$ is a **deterministic decoder** (0 learnable params).

Corollary 1.1

In a PaF model, the physics residual satisfies $r = O(\Delta t^p + \Delta x^q)$ for **all** $\theta \in \Theta$. The residual is at numerical discretization precision **regardless of network parameters**.

Taxonomy: Four Levels of Physics Incorporation

Level	Name	Physics How?	Residual $r(\theta)$	Example
L0	Pure Learning	None	Undefined	CycleGAN, U-Net
L1	Soft Physics (PINN)	Loss: λr^2	**Depends on** θ	PINNs, DiffUS
L2	Hard Constraint	Output projection	May depend on θ	Constrained layers

L3	Physics-as-Forward	Forward = Physics	Independent of θ	DPC-GNN (ours)

L1 (PINN)

$\frac{\partial r}{\partial \theta} \neq \mathbf{0}$

Physics is a *competing objective*

λ is a hyperparameter to tune

Physics satisfaction is approximate

L3 (Physics-as-Forward)

$\frac{\partial r}{\partial \theta} \approx \mathbf{0}$

Physics is *structurally guaranteed*

$\lambda = 0$ is the only correct choice

Physics satisfaction is exact

Theorem 1 (Physics-Weight Redundancy)

For any PaF model, consider $\mathcal{L}_\lambda(\theta) = \mathcal{L}_{\text{data}}(\theta) + \lambda \|r(\theta)\|^2$. Then:

1. $\|r(\theta)\| = O(\epsilon_{\text{machine}})$ for all θ (by Corollary 1.1)
2. $\nabla_\theta \|r(\theta)\|^2 = O(\epsilon^2_{\text{machine}})$ (numerical noise only)
3. Therefore $\nabla_\theta \mathcal{L}_\lambda \approx \nabla_\theta \mathcal{L}_{\text{data}}$ for all λ

Implication: $\lambda = 0$ is not a hyperparameter choice — it is an **architectural necessity**.

2. Cross-Domain DPC-GNN Transfer

Definition 2 (DPC-GNN Framework)

The DPC-GNN framework for physical domain $\mathcal{D}$ consists of:

1. **Constitutive Encoder** $\mathcal{E}_\theta$: Observable data → material parameters
2. **Physics Engine** $\mathcal{P}$: Deterministic solver (0 learnable params)
3. **Antisymmetric MP**: $\mathbf{W}_{\text{anti}} = \mathbf{W} - \mathbf{W}^\top$ guarantees conservation law
4. **Data-only Loss**: Supervision only on final observables

Symbol	Soft Tissue	Acoustic	General
$\phi(\mathbf{x})$	Lamé (μ, λ)	Acoustic (c, α, σ)	Constitutive params
$\mathbf{u}(\mathbf{x}, t)$	Displacement	Pressure p	State variable

$\mathcal{L}[\mathbf{u}]$	$\rho\ddot{\mathbf{u}} = \nabla \cdot \mathbf{P}$	$\ddot{p} = c^2\nabla^2 p$	Governing PDE
$\mathcal{C}$	$F_{ij} = -F_{ji}$	$m_{ij} = -m_{ji}$	Conservation/symmetry
$\mathcal{E}_\theta$	Geometry → (μ,λ)	CT → (c,α,σ)	Constitutive encoder
$\mathcal{P}$	Leapfrog	Leapfrog	Physics solver

Theorem 2 (Domain Transferability)

DPC-GNN transfers from domain $\mathcal{D}_1$ to $\mathcal{D}_2$ if:

1. $\mathcal{L}_2$ admits a stable explicit time-stepping scheme
2. $\mathcal{C}_2$ can be expressed as antisymmetry on the MP kernel
3. Constitutive parameters of $\mathcal{L}_2$ are predictable from observable data

Implication: DPC-GNN can extend to Maxwell's equations, heat conduction, and (with stabilization) Navier-Stokes.

3. Antisymmetric Message Passing

The core invariant shared across all DPC-GNN variants:

$$\mathbf{W}_{\text{anti}} = \mathbf{W} - \mathbf{W}^\top, \quad \mathbf{m}_{ij} = \tanh((\mathbf{h}_i - \mathbf{h}_j) \cdot \mathbf{W}_{\text{anti}})$$

Proposition 2 (Message Antisymmetry)

For any odd activation ϕ and antisymmetric kernel $\mathbf{W}_{\text{anti}} = -\mathbf{W}_{\text{anti}}^\top$:

$$\mathbf{m}_{ji} = \phi((\mathbf{h}_j - \mathbf{h}_i) \cdot \mathbf{W}_{\text{anti}}) = \phi(-(\mathbf{h}_i - \mathbf{h}_j) \cdot \mathbf{W}_{\text{anti}}) = -\phi((\mathbf{h}_i - \mathbf{h}_j) \cdot \mathbf{W}_{\text{anti}}) = -\mathbf{m}_{ij}$$

Domain	m_{ij} represents	$m_{ij} = -m_{ji}$ enforces	Physical basis
Soft Tissue	Inter-node force	Newton's 3rd law	Momentum conservation
Acoustic	Spatial influence on params	Spatial reciprocity	$c^2\nabla^2$ is self-adjoint

4. Differentiable Wave Propagation

Leapfrog as a Differentiable Map

Each time step defines $\mathcal{T} : (p^{n-1}, p^n) \mapsto p^{n+1}$:

$$p^{n+1} = \frac{2p^n - (1 - \alpha\frac{\Delta t}{2})p^{n-1} + \Delta t^2(c^2\nabla_h^2 p^n + \sigma s^n)}{1 + \alpha\frac{\Delta t}{2}}$$

Jacobian w.r.t. Sound Speed

$$\frac{\partial p_{i,j}^{n+1}}{\partial c_{k,l}} = \frac{2\Delta t^2\, c_{k,l}\, \nabla_h^2 p_{k,l}^n}{1 + \alpha_{i,j}\frac{\Delta t}{2}} \cdot \delta_{ik}\delta_{jl}$$

The Jacobian is **diagonal** (local) — computationally favorable.

> **Proposition 3 (Gradient Boundedness)**
>
> Under CFL condition, the spectral radius of the step Jacobian satisfies:
>
> $$\rho\left(\frac{\partial \mathcal{T}}{\partial p}\right) \leq \frac{2 + 4\,\mathrm{CFL}^2}{1 + \alpha_{\min}\frac{\Delta t}{2}}$$
>
> For our parameters (CFL = 0.206): $\rho \leq 2.170$. With attenuation $\alpha > 0$, the effective radius is further damped.

End-to-End Gradient Path

$$\frac{\partial \mathcal{L}}{\partial \theta} = \underbrace{\frac{\partial \mathcal{L}}{\partial I_{US}}}_{\text{B-mode loss}} \cdot \underbrace{\frac{\partial I_{US}}{\partial \mathbf{s}}}_{\text{DAS}} \cdot \underbrace{\frac{\partial \mathbf{s}}{\partial p^N}}_{\text{sensors}} \cdot \underbrace{\frac{\partial p^N}{\partial(c, \alpha, \sigma)}}_{\text{Leapfrog chain}} \cdot \underbrace{\frac{\partial(c, \alpha, \sigma)}{\partial \theta}}_{\text{GNN}}$$

5. Physics-Prior Residual Learning

Sound speed is decomposed into known physics and learned residual:

$$c_\theta(\mathbf{x}) = \underbrace{c_{\text{table}}(\mathrm{HU}(\mathbf{x}))}_{\text{known prior}} + \underbrace{\Delta c_\theta(\mathbf{x})}_{\text{GNN residual}}$$

Proposition 4 (Information Decomposition)

$$I(\mathrm{CT}; c^*) = I(\mathrm{CT}; c_{\mathrm{table}}) + I(\mathrm{CT}; c^* \mid c_{\mathrm{table}})$$

c_{table} captures the dominant mode (tissue-type mapping). The residual Δc captures:

1. **Spatial context**: tissue boundary interactions
2. **Sub-resolution structure**: below CT resolution, above US wavelength
3. **Patient-specific variation**: deviations from population averages

Initialization benefit: With $\Delta c_\theta^{(0)} \approx 0$, the initial forward pass uses $c \approx c_{\mathrm{table}}$, producing physically plausible waves from epoch 0.

6. Computational Complexity

Component	FLOPs / sample	Memory	Learnable?
GNN encoder (253K)	~500K	~2 MB	✅
Leapfrog (200 steps)	~65M	~5 MB (ckpt)	❌
DAS beamformer	~21M	~8 MB	❌
Total	**~86M**	**~15 MB**	

Method	Time/sample	Speedup	Physics guarantee
k-Wave (256^2, OMP)	3,200 ms	1×	✅ Full
DPC-GNN-Acoustic V4	**< 100 ms**	**> 32×**	✅ **Full**

7. Summary of Contributions

#	Contribution	Key Result
C1	**Physics-as-Forward formalization**	Definition 1 + Theorem 1: $\lambda = 0$ is architectural necessity

C2	**Cross-domain DPC-GNN transfer**	Theorem 2: solid mechanics → wave propagation
C3	**Antisymmetric MP generalization**	Proposition 2: force balance → spatial reciprocity
C4	**Differentiable acoustic simulation**	Proposition 3: gradient stability through 200 Leapfrog steps
C5	**Practical CT-to-US system**	253K params, <100ms, guaranteed physics

"Physics is not a loss. Physics is the forward pass."